

일반물리II (BS105a-01)

학기	2026년 2학기
담당교수	DEVECIOGLU DENIZ OLGU
이수구분	공통필수
학점	3.0
강의시간	월11:00-12:30(E7 - 241), 수11:00-12:30(E7 - 241)
강의실	
소속	기초학부

교과개요

This course introduces the fundamental concepts of electromagnetism in an accessible and intuitive manner. Topics include electric forces and fields, electric potential, electric current, resistance, and basic electrical circuits. Students will explore magnetic fields, magnetic forces, and electromagnetic induction, with emphasis on practical applications. The course concludes with an introduction to electromagnetic waves, highlighting their properties, behavior, and role in the transmission of energy.

교과목표

By the end of the course, students will be able to:

- Apply vector analysis techniques to problems in electromagnetism
- Explain key principles of electrostatic and magnetostatic fields
- Demonstrate problem-solving skills in various electromagnetic contexts
- Build a strong foundation for further study and practical applications in electromagnetic theory and engineering

주차별 강의계획

1주차: Static Electricity; Electric Charge and its Conservation; Insulators and Conductors; Induced Charge; Coulomb's Law; The Electric Field

2주차: Electric Field of a Continuous Charge Distribution; Electric Field Lines; Electric Fields and conductors; Motion of a Charged Particle in an Electric Field; Electric Dipoles

3주차: Electric Potential Energy and Potential Difference; Relation Between Electric Potential and Electric Field; Electric Potential due to Point Charges; Potential Due to Any Charge Distribution; Equipotential Lines and Surfaces; Applications of Electrostatics

4주차: Definition of Capacitance; Calculation of Capacitance; Capacitors in Series and Parallel; Storage of Electrical Energy; Dielectrics; An Atomic Description of Dielectrics

5주차: Electric Current; Ohm's Law: Resistance and Resistors; Resistivity; Model for Electrical Conduction; Resistance and Temperature; Superconductors; Electrical Power

6주차: Midterm

7주차: EMF and Terminal Voltage; Resistors in Series and Parallel; Kirchhoff's Rules; EMFs in Series and Parallel; Charging a Battery; RC Circuits – Resistor and Capacitor in Series

8주차: Magnets and Magnetic Fields; Electric Current Produce Magnetic Fields; Force on an Electric Current in a Magnetic Field: Definition of B; Force on an Electric Charge Moving in a Magnetic Field; Torque on a Current Loop

9주차: Magnetic Dipole Moment; Applications: Motors, Loudspeakers, Galvanometers; The Hall Effect

10주차: The Biot-Savart Law; The Magnetic Force Between Two Parallel Conductors; Ampere's Law; The

Magnetic Field of a Solenoid; Gauss's Law in Magnetism; Magnetism in Matter

11주차: Induced EMF; Faraday's Law of Induction; Lenz's Law; EMF Induced in a Moving Conductor; Electric Generators; Back EMF and Counter Torque; Eddy Currents; Transformers and Transmission of Power; A Changing Magnetic Field Produces an electric Field

12주차: Mutual Inductance; Self-Inductance; Inductors; Energy Stored in a Magnetic Field; LR Circuits; LC Circuits and Electromagnetic Oscillations; LC Oscillations with Resistance (RLC Circuit)

13주차: AC Circuits and Reactance; RLC Series AC Circuit; Phasor Diagrams; Resonance in AC Circuits

14주차: Changing Electric Fields Produce Magnetic Fields; Displacement Current; Gauss's Law for Magnetism; Maxwell's Equations

15주차: Production of Electromagnetic Waves; Electromagnetic Waves, and Their Speed, Derived From Maxwell's Equations

16주차: Finals

학습윤리

You are expected to demonstrate academic honesty in all aspects of this course. Academic dishonesty includes, but is not limited to: violating clearly stated rules for taking an exam or completing an assignment; plagiarism (including material from sources without a citation and quotation marks around any borrowed words); claiming another's work or a modification of another's work as one's own; buying or attempting to buy papers or projects for a course; fabricating information or citations.

교과정책(방침)

Attendance is mandatory and will be formally graded. Assignments and exams must be submitted on time. Late submissions will result in point deductions that may affect your final grade. If you anticipate a conflict with a due date or encounter difficulties with an assignment, you must plan ahead and notify the instructor in advance. This will allow us to discuss the situation and, if appropriate, make alternative arrangements.